
IMPACTS OF 21ST CENTURY SEA LEVEL RISE ON TROPICAL COASTAL ECOSYSTEMS



Marine Conservation Philippines

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Introduction

Sea level rise is one of the most imminent threats to coastal communities, it's interaction with the coastline defines it. Over the last 150 years the accuracy of which sea level can be measured by has much improved. Now the records can provide us with an accurate picture of changing sea level on global and regional scales. This review will explore how it may affect different tropical coastal ecosystems in the coming decades. The evidence that points to coastal ecosystems suffering increasingly adverse impacts to sea level rise and its attributed consequence have been quoted with very high confidence (Wong et al., 2014). As they support so many people it is very important that we can understand the limits of them.

To best mitigate impacts of sea level rise on coastal ecosystems, the aims of this will be to assess:

1. What is the average rate of global sea level rise?
2. What is the rate of sea level rise in the Philippines?
3. What defines a tropical coastal ecosystem?
4. What are the stresses and responses of these ecosystems to sea level rise?
5. How vulnerable is Siit Bay to 21st century sea level rise?

Siit Bay in Negros Oriental Philippines will be taken as a case study and tested against criteria to observe how well it will adapt the projected changes in sea level rise. This provides us with an excellent opportunity to test the limits of how mangroves can adapt to changes in sea level rise. It is an exciting opportunity to quantify the resilience of this coastal ecosystem.

Sea Level Rise

The most comprehensive research conducted on sea level rise is published by the International Panel on Climate Change (IPCC). They review and combine literature to estimate historic sea level, measure instrumental records, and model projections for the future. This is shown in Figure 1, and displays how sea level does not begin to rise until the post-industrial era. Following this the instrumental record starts and sea level rises. The projections for the future are displayed based on different socio-economic scenarios that change greenhouse gas emissions and radiative forcing. These are called RCP Scenarios (Representative Concentration Pathways). The fourth edition of the report released in 2007 estimated that sea level rise would increase on average by up to 60cm by 2100 (Solomon S et al., 2007), with potential scenarios that factor in accelerated ice sheet melting increasing this to 80cm (Pfeffer et al., 2008). This estimate was revised to up to 74cm in a RCP8.5 scenario in the next edition of the IPCC reports, which assumes unmitigated rises in future emissions (Wong et al., 2014).

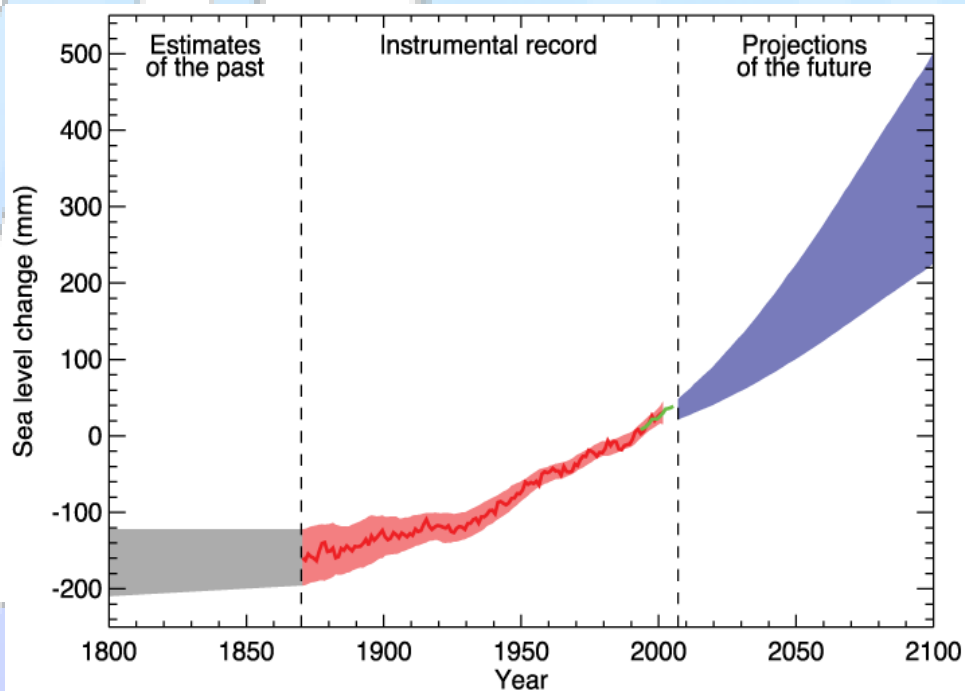


Figure 1 – Sea level estimations, records and projections from 1800 to 2100 (Wong et al., 2014).

There are many different components that contribute to the rise or fall of sea level. (Rietbroek et al., 2016) explored the global and regional contributions to sea level change using data generated from the Gravity Recovery and Climate Experiment (GRACE) which records gravity anomalies combined with Jason-1/2 data, a project recording sea level anomalies. From this, sea level changes can be calculated at global and regional scales with the resolution of the component included.

The different components are steric (volumetric changes due to temperature and salinity), ice sheet melting (Greenland, Antarctica and other land ice), glacial isostatic adjustment (GIA) caused by tectonic adjustments, hydrology mass variability and other stresses including longer term atmospheric patterns of wind and heat/freshwater fluxes.

As shown in Figure 2, the findings of the report are globally that the steric contribution is larger (1.38 ± 0.16 mm/y) than previous models derived from in-situ temperature and salinity profiles (ranging from 0.66 ± 0.20 mm/y to 0.96 ± 0.10 mm/y).

Greenland is providing an increasingly large amount of freshwater with contributions increasing from 3 to 9 mm per year, this links back to papers that are arguing that the increased amount of melting ice sheets will provide larger than expected sea level rises than current projections calculate (Pfeffer et al., 2008), (Nicholls et al., 2010).

The one component providing a negative contribution is hydrology. This could possibly point to processes of damming which holds up water on land before it reaches the ocean.

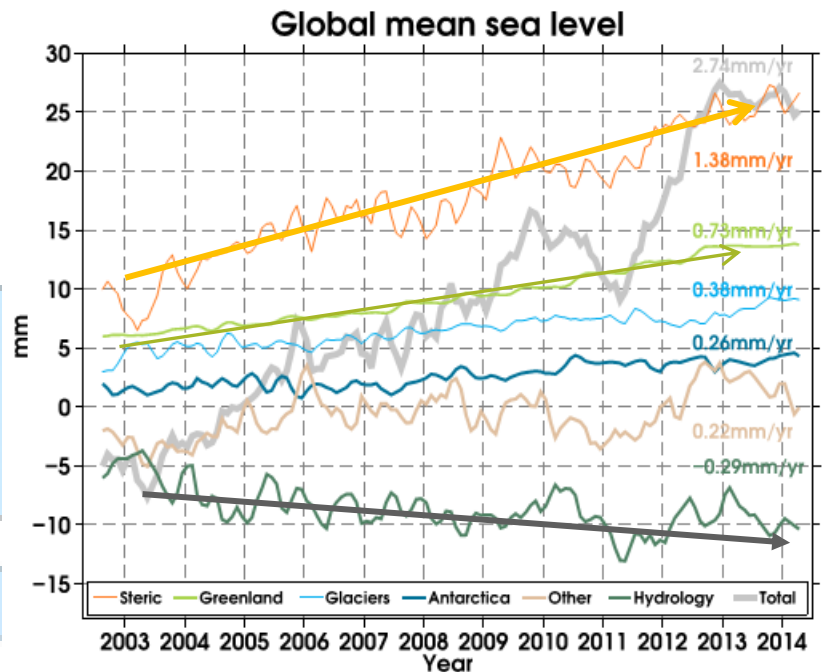


Figure 2 - The different components of sea level change (Rietbroek et al., 2016)

Philippines Sea Level Rise

As discussed in the introduction, the Philippines is vulnerable to climate change and sea level rise. (Rietbroek et al., 2016) found that it is the Philippines and the waters to the east of the country that have had the highest increase of sea level with 14.7mm/year. In Figure 3, a red box highlights the Philippines study area. It shows that approximately 75% of this increase is due to steric expansion.

It is a widely documented fact, with numerous sources quoting the Philippines to have 3 or 5 times the global average. The warmer water correlated with more frequent and intense typhoons which has all contributed to its vulnerability. The Global Climate Risk Index study in 2016 placed the Philippines ninth from 1993 – 2005 and first in 2013 as most vulnerable (Kreft et al., 2016).

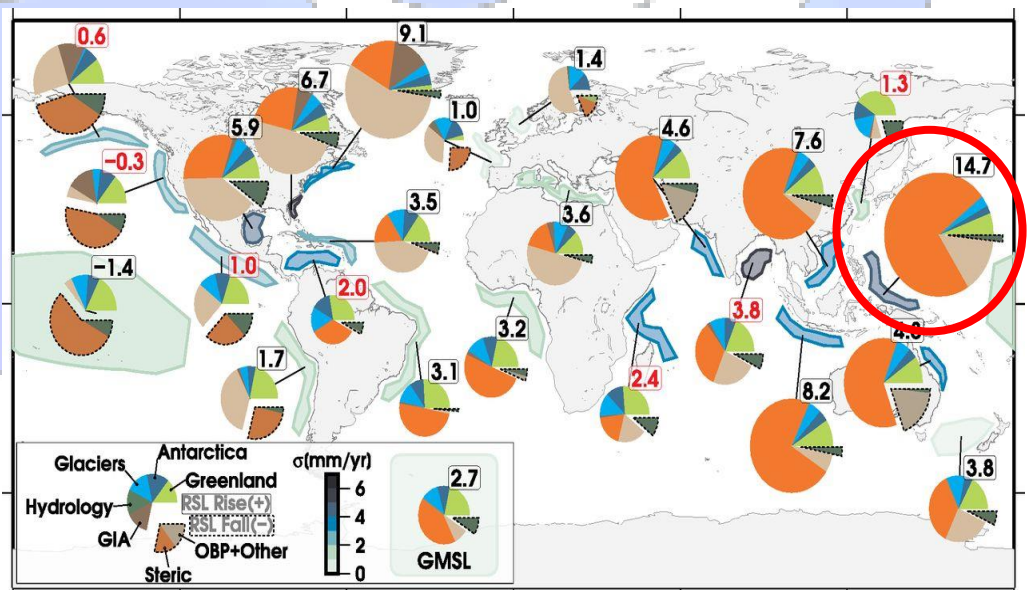


Figure 3 - RIETBROEK ET AL 2016 STUDIES RELATIVE SEA LEVEL RISE IN SELECTED COASTAL LOCATIONS GLOBALLY.

When the different components of sea level record are cross-examined with ENSO (El Niño Southern Oscillation – the scale which defines whether there is an El Niño or La Niña phase) there is a direct correlation of the following, as shown in Figure 4.

1. El Niño – Lower sea level as trade winds weaken and water pooled up by equatorial winds moves towards the eastern Pacific
2. La Niña – Extreme of normal conditions where water is warmer than usual and pushed towards the western Pacific.

Interestingly the waters in the central Pacific have fallen, and this may suggest this water is being pooled up in the west Pacific.

As well as highlighting the importance of natural phenomena in sea level rise, this highlights how ideally a longer study time would be ideal when trying to determine future trends at region resolution. It also breaks the increases down as fluctuations. This concept of rapid changes due to events such as El Niño and La Niña is something that locally the ecosystems have adapted to historically as they are present and functioning today.

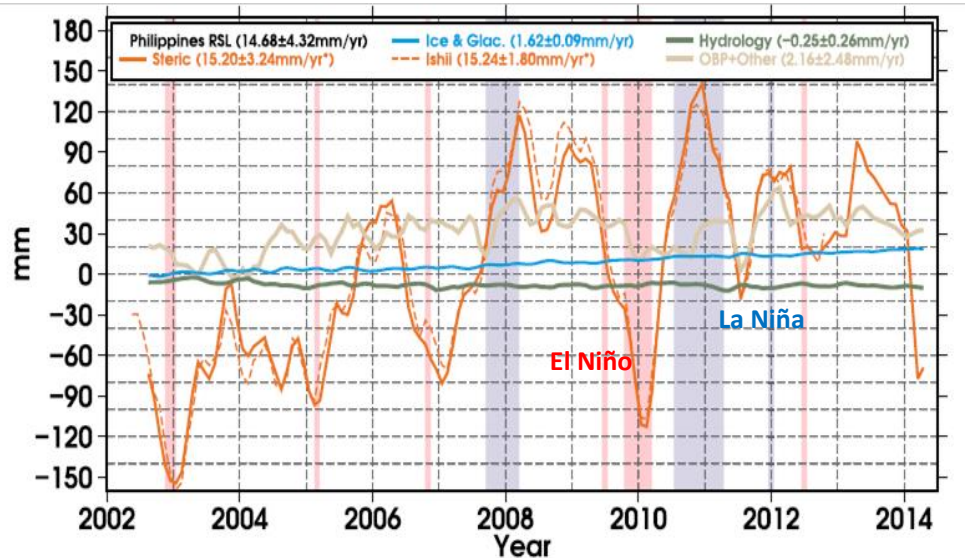


Figure 4 - Correlating sea level rise components against ENSO cycles. (Rietbroek et al., 2016).

Coastal Ecosystems

Coastal ecosystems can be defined as coastal lands where fresh and salt water mix, and associated near shore marine areas (MEA, 2001). In the tropics, this usually consists of a transition of mangroves on land and in the inter-tidal area, to sea grass in the shallow sandy substrate mixing in as a mosaic with coral reefs until coral dominates to its seaward limit, where the conditions cease to meet their requirements (Burke et al., 2001).

To explain how climate change directly affects these coastal ecosystems we must understand how they work and would be perturbed by the pressures of sea level rise. It has been quantified for the first time by (Saunders et al., 2014) on the Great Barrier Reef, Australia. The model found scenarios where the increased depth of water would increase the exposure of seagrass and associated nursery grounds to increased wave energy, moving their seaward edge towards land. Regarding mangroves, studies have been done to assess the past rates of which mangroves have adapted to sea level rise. The likelihood of adaptation depends on its environmental conditions and these will be referenced against Siit Bay to assess its vulnerability.

Coral Reefs

Coral reefs maintain some of the most biodiverse areas on the planet. They are adapted to living in a wide range of conditions such as fresh or salt water and warm or cold temperatures. In tropical coastal areas, they work to protect the nursery grounds within seagrass areas (Peevor & Carey, 2009). Projections of sea level rise globally stand at 3mm on average with previously discussed rates of 14mm per year in the Philippines. It is widely reported in literature that coral can keep pace with sea level fluctuations (healthy flat reefs accrete at 3mm/year

(Buddemeier & Smith, 1988) and even promote reef growth vertically (Wilkinson & Buddmeier, 1994), including slow growth reefs in a scenario where this is the only factor (Edwards, 1995). However, the reality of rising seas implicates an increasing ocean temperature which would itself reduce the rate at which the coral can grow at. Research by (Yates et al., 2014) has quoted that more than 90% of all reefs will be threatened by 2030 due to climate change. The specialized way in which corals survive and grow is directly impacted by how the increasing amount of carbon dioxide in the atmosphere affects ocean chemistry.

Seagrass Beds

Seagrass beds provide the buffer between the coral reefs and the mangroves. They are also important as they connect the mangrove environment to the reefs. In addition, they are key blue-carbon ecosystems, with the amount of CO₂ absorbed per hectare on the same lines as that sequestered by the Amazon rainforest (Saunders, 2013)

Figure 5 is a graphic on how the coral shelters the seagrass from increased wave energy, and how an increasing water column will change the wave energy getting to more protected areas of the reef. The wave orbitals (the motion of how energy generated by waves is directed in water) is show to increase in strength with rising sea.

As sea level increases, the area at which light penetrates to decreases reducing the amount the seagrass can photosynthesize. Research done in Queensland has shown that through good management, increasing the water clarity by 30% can offset the impacts of 1.1m of sea level rise (Saunders, 2013).

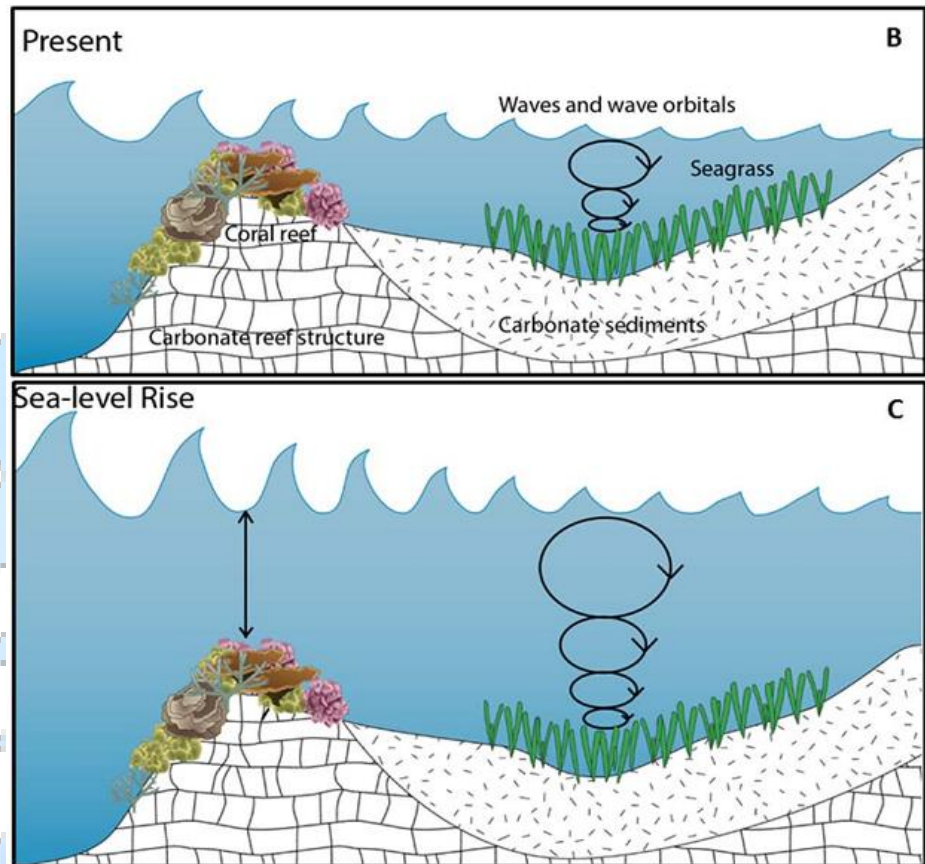


Figure 5 - The interaction of coral reefs and seagrass beds in responses to rising sea level (Saunders et al., 2014)

Seagrass in a calmer environment than the reef will further act to dissipate the wave energy through extra friction. This regulates the quantity of sediment transported to the mangroves and allows the mangrove seedlings to establish in the slacker waters.

Mangroves

The previously mentioned impacts of sea level rise on coral reefs and seagrass beds impact the mangrove ecosystem. As the mangrove lies in the intertidal zone, a change in sea level directly and immediately impacts the ecosystem. The Philippines is blessed with some of the most biodiverse sets of mangrove plants along with other south east Asian countries such as Indonesia with 40-50 of the 76 true mangrove species (Ashraf & Habjoka, 2013).

During the last 100 years, there has been an increasing population in the Philippines from ~7.6 million in 1900 to nearly 100 million today. This has led to the reduction in area of mangroves from 450,000 in 1920 to 117,000 hectares currently (Primavera, 2000). The relationship between population increase and mangrove forest area is shown in Figure 6 in a study assessing institutional issues and the conservation of mangroves in the Philippines (Primavera, 1995). This is due to many reasons, for example, space for development, firewood, aquaculture and firewood (Melena, 2000).

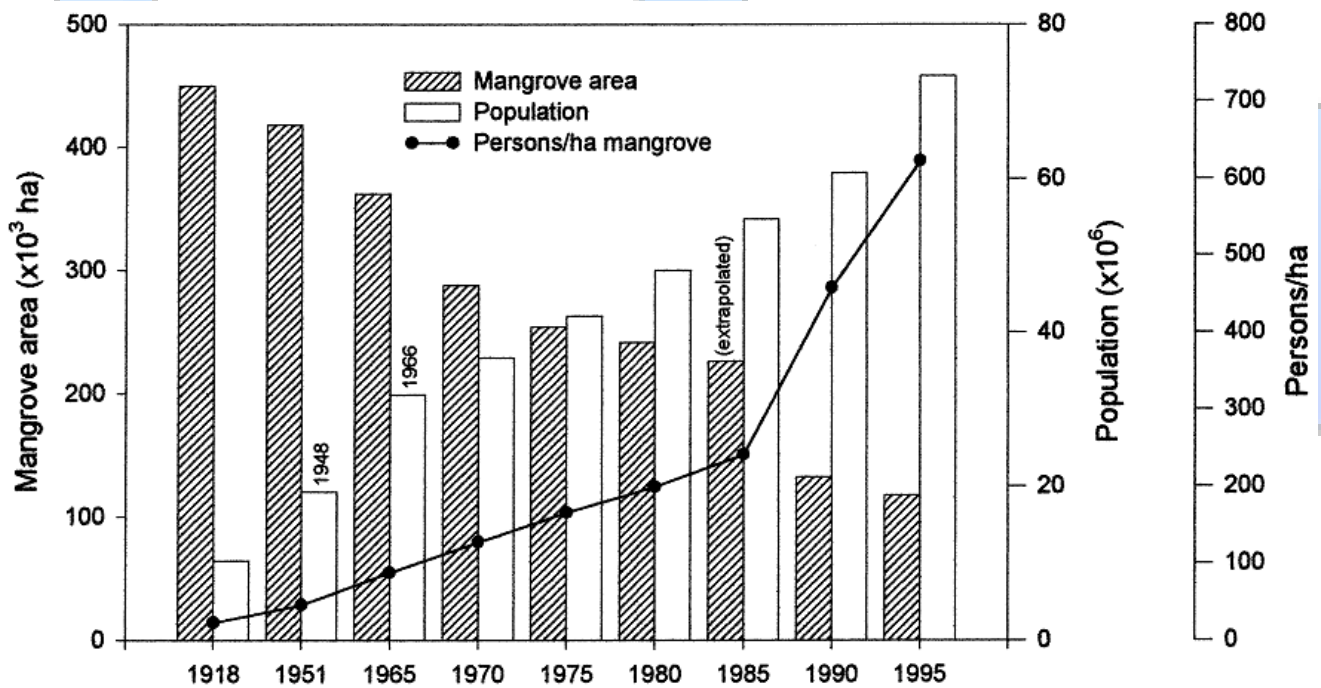


Figure 6 - Relationship between population and mangrove area (Primavera, 1995)

Contrary to human impacts and development exploiting the mangrove ecosystem, the next 100 years the mangroves will be mostly threatened for natural reasons. As our awareness grows for the protection and conservation of these ecosystems, the human impact will optimistically be to increase the health and area of mangroves.

Out of the different impacts of climate change, it is sea level rise that will impact the mangroves mostly. This is due to their location in the intertidal zone, where species are specifically adapted to tolerate different

hydroperiods and salt concentrations. The range of species and their position in the mangrove ecosystem is shown in Figure 7. Mangroves are fine-tuned to living in an inter tidal regime through many specialties such as pneumatophore roots in plants such as *Sonneratia* and *Avicennia* (Ellison & Farnsworth, 1997). These are roots which grow vertically upwards to get air in anoxic conditions. Another adaptation is compartmentalizing; this involves branches directing toxins and salts towards one leaf in the branch which is sacrificed so the others can survive. This resilience has enabled them to thrive for the last 65 million years, and adapt to changes in sea level of over 100m between glacial periods.

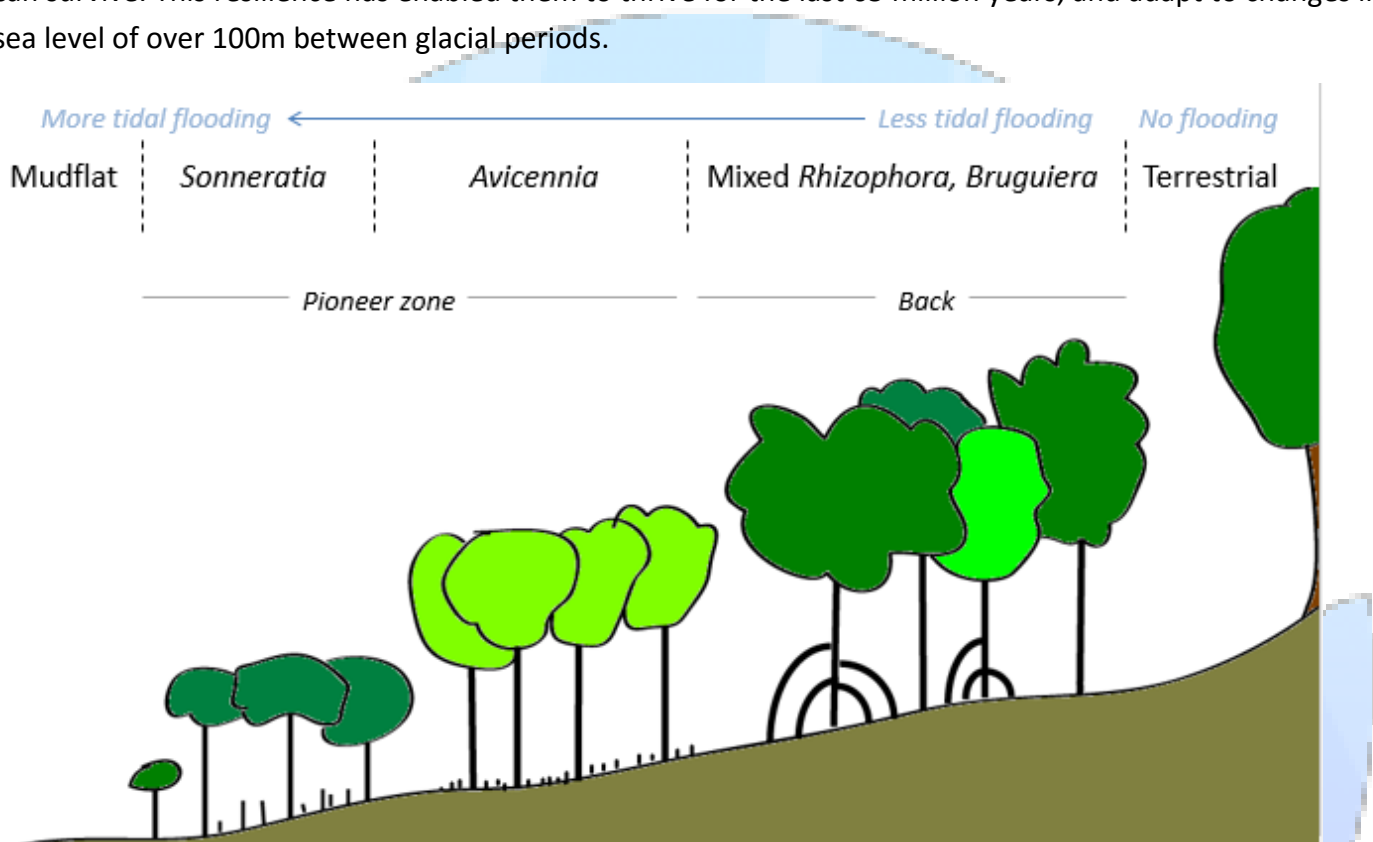


Figure 7 - Species location within a mangrove ecosystem (MacKenzie & Fries, 2015)

Having survived sea level swings before, it is known the mangrove ecosystem is well equipped to adapt to these changes. There are different ways in which the mangrove can adapt to rising sea level.

Marsh Accretion

This is the primary response for the mangrove. Studies show (Alongi, 2008) that in a 1:1 ratio of sea level rise and accretion rates, most mangroves are accreting at a rate the same or greater than sea level rise. The three key factors that contribute to the rising of the mangrove forests include the following (MacKenzie & Fries, 2015).

- Root growth in sediment – this requirement is the growth of small roots that can trap and establish the sediment that is travelling through the mangrove. As it establishes it raises the level of the mangrove.
- Sediment input – a balanced sediment input which includes both inorganic and organic matter from marine and upland sources.

- Leaf litter input – this process works to quickly return organic nutrients to the forest floor through the influence of consumption or burial by organisms such as crabs.

If the rate of sea level exceeds that of accretion, then the mangrove may start to become submerged at its seaward edge. This does not just have to depend on sea level, but a few other factors and link to the conditions above. The main effects of sea level rise on the surface elevation changes (how the mangrove may accrete or subside) in a negative way are (McIvor, 2013);

- Increased time inundated in water
- Increased wave energy – resuspension and erosion of sediment
- Water logging – anoxia (areas of zero oxygen)
- Rise of groundwater level - saline intrusion

Landward Migration

When stressed, mangroves also have the capacity to migrate landward. Even if the mangrove begins to be submerged at its seaward edge, there are some factors that can help towards the landward migration of the mangrove. They include;

- High species diversity – increased chance of having species with characteristics of pioneering new zones and establishing sediment to equal sea level rise
- Absence of land barriers – it is crucial that there is adequate room for the mangroves to move in. Roads and seawalls are good examples of common barriers, but they also include topography. This would be where the mangrove backed up on to a steep mountain or hill.

This scenario would usually be applied to more remote settings where the natural mangrove is in place without the presence of a squeeze to the developed areas.

The range of geomorphic settings (types of beach/lagoon/land) are an important factor on the likelihood of successful adaptation. Specific conditions found locally in Siit Bay need to be examined to determine its viability for long term projects.

Case Study: Siit Bay, Negros



The bay is located near the southernmost tip of Negros, in the middle belt of the Philippines, the Visayas. During this paper, many different factors to how stressed a coastal ecosystem will be to sea level changes are highlighted. To categorize how vulnerable Siit Bay is to future sea level increase, it can be cross referenced against characteristics published by (McIvor, 2013) in Table 1. To summarize, the key factors behind vulnerability have been taken from this and answered.

- Relief – NOT VULNERABLE - The island has a volcanic history. It is well established and at no risk of subsidence.
- Geomorphology – NOT VULNERABLE - The mangrove is located behind a spit that runs parallel to shore. This protects the mangrove from increased wave energy. However, in typhoons it is known that the water breaches this and floods the mangrove.
- Tidal Range – VULNERABLE - The maximum tidal range is 2.15m (closest tide station Dumaguete ~35km from site). This classifies as a meso-tidal environment where a micro tidal (vulnerable) area is < 2m.
- Sediment Input – NOT VULNERABLE - There is no riverine input to the mangrove. There are many creeks and a large catchment area. The annual rainfall
- Biodiversity – NOT VULNERABLE - The Philippines has one of the highest species diversity globally. It is estimated the species diversity in the bay is between 40 and 50 of the 76 true mangrove species.
- Bedrock activity – NOT VULNERABLE - Middle of Philippine plate with little risk of major tectonic activity. No commercial groundwater activity.
- Development – NOT VULNERABLE - Relatively remote setting. One coastal road that is constructed above the height where mangroves can migrate to. Small settlements around the bay with small population with ~400m of space to migrate landward.

Table 1 - Defining the vulnerability of a mangrove ecosystem to climate change (McLeod & Salm, 2006)

Vulnerability	Local Conditions	Explanation
Most Vulnerable	Low relief islands	- low rates of sediment and peat accretion, particularly vulnerable to sea-level rise because they are subject to drought and wave erosion - expected to experience increased flooding, inundation and salinization of soils and freshwater (Shea et al. 2001).
	Lack of rivers	- lack of sediment and freshwater
	Carbonate settings	- often associated with atolls and islands, where landward migration to escape sea-level rise may not be possible - sediments are mostly locally derived
	Areas subsiding due to tectonic movements, groundwater extraction, or underground mining	- will experience higher sea-level rise and inundation
	Micro-tidal sediment starved environments (small Caribbean islands) (Ellison 1993)	- lack of sediment will lead to decreased geographic distribution and species diversity of mangroves (Houghton et al. 2001)
	Mangroves blocked by coastal development or steep topography	- unable to move inland when sea level rises
Least Vulnerable	Mangroves in deep sediment on high islands	- structurally stronger than mangroves in shallow sediment on low islands (Gillison 1980) and less vulnerable to storm surges than low islands (UNEP 1994) - high islands will be better adapted to survive predicted climate changes due to their larger surface areas, freshwater availability, better soils, and more diverse resources (Shea et al. 2001).
	Riverine mangroves	- receive large amounts of sediment from other areas (Woodroffe and Grindrod 1991) - most productive mangrove habitats due to high nutrient concentrations associated with sediment trapping (Ewel et al. 1998).
	Macro-tidal sediment rich environments (mangroves in northern Australia)	- access to sediment and strong tidal currents to redistribute sediment (Woodroffe and Grindrod 1991)
	Mangroves with room to move landward (backed by low-lying areas, salt flats, undeveloped areas)	- have the opportunity to expand inland when sea level rises
	Mangroves in remote areas	- have limited anthropogenic stresses and not blocked by coastal communities from moving landward
	Mangroves surrounded by flourishing dense mangrove forests	- have steady supply of propagules and seeds

The outlook for Siit from the criteria looks positive. Earlier, we assessed that the Philippines has the highest rate of sea level rise globally. To combat this, the mangroves need the best set of characteristics to combat and adapt to it. Huge advantages are its remote setting, guarding from a sand barrier and high species diversity. The room of about 400m to migrate towards is adequate and available for colonization.

Suggestions for further work

Accurate mapping of the entire of Siit Bay coral seagrass and mangrove zones, combined with surface elevation measurements and accurate sea level measurements. This would provide a holistic view of the tropical coastal ecosystems response to 21st century climate change. Current studies and restoration programmers are being conducted by organizations such as the Zoological Society of London (ZSL) who are in Bohol and Panay incorporating MPAs into their restoration methods. Possibly, there could be research done into the coral and seagrass health around Siit to see what current condition they are in.

Other successful ventures have been to introduce eco-tourism using boardwalks and information boards to generate money to local communities. This could incorporate reintroduction of native species of birds and animals not currently living in the area.

Detailed studies using high resolution technology have primarily been used in wealthier countries such as the USA and Australia, and a detailed study here could potentially give an insight to mangroves in the Philippines and Indonesia that are distinctly different from those in other countries.

Summary

The Philippines location and geography subjects itself to the brunt of sea level rise. Whereas the global average is +3.26 mm per year, it is >14 mm per year due to mostly steric expansion. There are possibilities that this could link to changing ENSO cycles as the trade winds blow the warm water across to the western margin of the Pacific. As well as other effects of this change – such as increased frequency and intensity of typhoons we have learnt that it will impact the coastal ecosystems. The 60% of Philippine people living within the coastal zone rely on its uses and need to preserve and maintain what is left.

The tropical coastal ecosystems that are seen in the Philippines are mostly composed of coral reefs, seagrass beds and mangrove forests. A huge percent of these have degraded or been demolished because of development over the past 100 years. These ecosystems are interconnected and rely on each other to survive through feedbacks. Research shows whilst sea level rise may not impact the coral reefs and seagrass beds the most of climate change, it will impact on water clarity reducing and wave energy increasing. Mangroves however are most at threat of sea level rise. They are immediately impacted by the changing hydroperiod and rely their surface elevation to match that of sea level or migrate landward through colonization of new areas. In absence of this the ecosystem can break down.

In the example study, by looking at criteria that can define how vulnerable the system is to sea level rise, it is decided that the bay is under relatively low threat. This is due to its remote setting, high biodiversity and remote location allowing it to migrate up to 400m away from the coast. In addition to this the promotion of conservation and establishing of nursery grounds will further help to reinforce the long-term future of the bay. For these reasons, it is an ideal location to research with as little high resolution study has been done in these types of locations.

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